OVIZIO Challenge – Cell parameters

You have been hired to write an application to help Ovizio's data scientics to **visualize cells parameters** from captures taken with a Ovizio qMod system (hologram camera allowing to record images of a cell culture.

Note: Cell parameters are values such as diameter, circularity, optical height of cells.

User Requirements

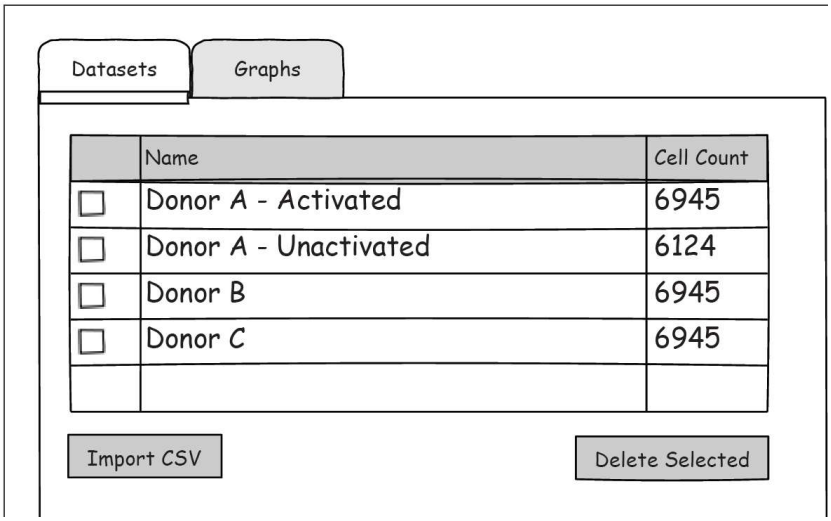
- The user can import a dataset from a csv file. These datasets will be persisted in the application.
- The user can delete datasets.
- The application will show the scatter plot of two parameters selected by the user. A different color will be used for each dataset.
- The application will show the histogram of one parameter selected by the user. A different color will be used for each dataset.

Technical notes

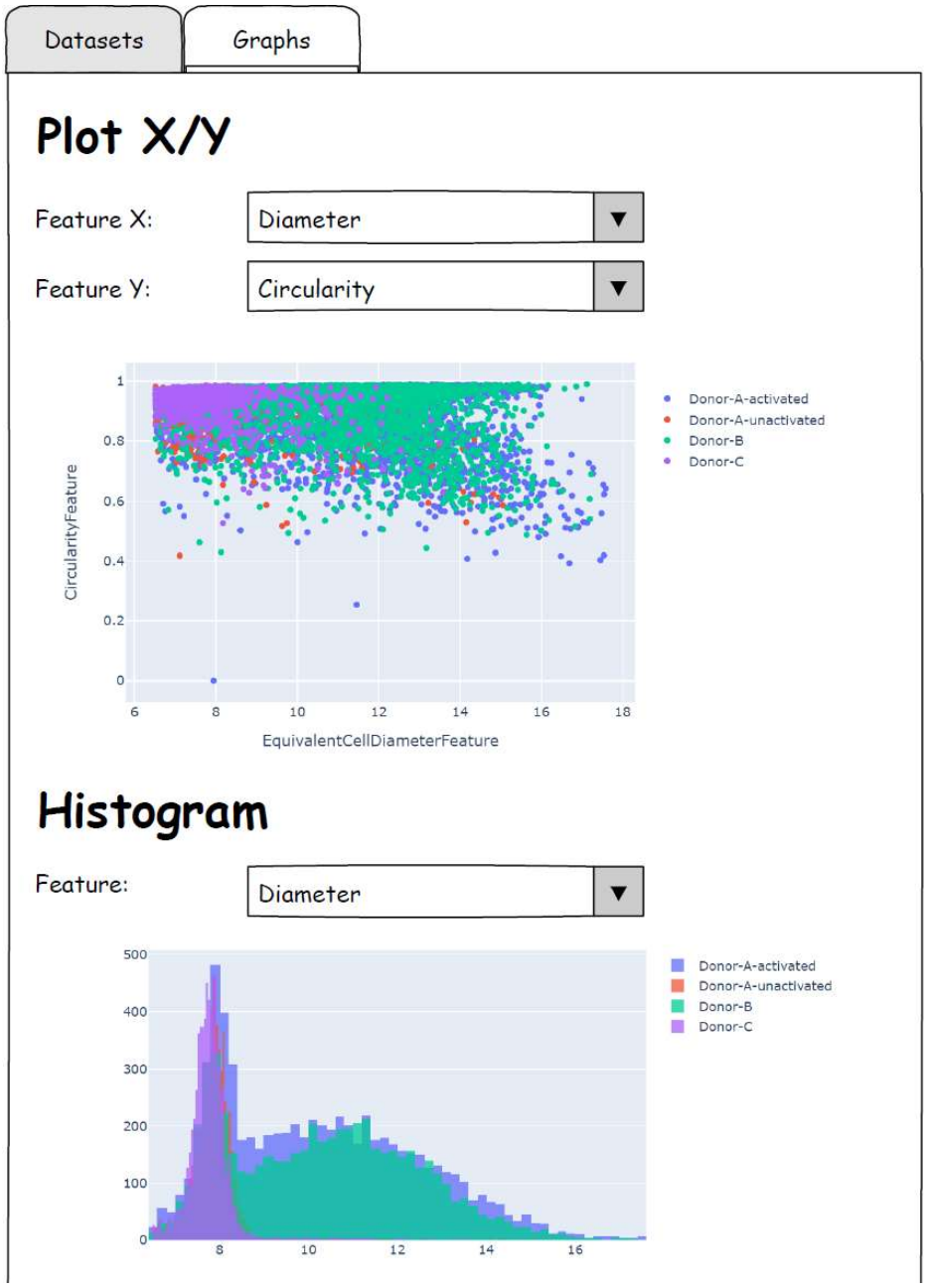
- The CSV are generated by Ovizio's QMod application . See the 4 provided files. It can be read easily with the pandas.read_csv function.
- The application will be written in python with the dash library. A tutorial can be found at https://dash.plotly.com/layout
- Data must be store in a SQL database (sqlite backend) with SQL . Here is an example how to store a dataframe in sql: https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.to_sql.html
- The application will be running in a container (nice to have)
- The application will implement a health check (see https://koalatea.io/python-health-check/, nice to have)

Mockups

Dataset screen



Graph screen



Enjoy coding! 😊